Estimating the Hubble Constant: From Historical to Modern Data

Alessandro Buttazzo
Independent research project, not conducted within a formal school or institutional research program

September 2, 2026
Revised November 2026 (version 1.3.0)

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Abstract

This project estimates the Hubble constant H_0 using two independent datasets: Edwin Hubble's original 1929 sample of 24 extragalactic objects, and a modern sample of 39 galaxies with distances and redshifts drawn from the NASA/IPAC Extragalactic Database (NED) and, for the six galaxies with geometric maser distances, from the Megamaser Cosmology Project. Refitting Hubble's data with his own model, which included the motion of the Sun, reproduces his published value of 465 km/s/Mpc; the comparison with the modern sample shows how limitations in the extragalactic distance scale of the 1920s led to an overestimate of the expansion rate by a factor of about six. For the modern sample, all distances are first converted to comoving distance, since the maser distances are angular-diameter distances and the others luminosity distances: although no galaxy is beyond $z = 0.034$, this changes the result by about 2 km/s/Mpc, more than its statistical error. The reference fit gives $H_0 = 72.5 \pm 1.5$ km/s/Mpc, with jackknife and bootstrap cross-checks. A sensitivity analysis shows that the six maser galaxies reduce the uncertainty by a factor of 2.7 and lower the central value by an amount within the uncertainty of the maser-free fit. An inverse-variance weighted fit illustrates a common pitfall (weighting by distance precision alone reintroduces the very peculiar-motion bias the sample extension was designed to suppress) and is repeated with a peculiar-velocity scatter estimated from the data; an orthogonal distance regression additionally allows for measurement error on distance, and a Monte Carlo simulation shows that all three estimators are unbiased to within 0.3 km/s/Mpc. A paired bootstrap shows that the spread among the ten resulting estimators (69.6–74.9 km/s/Mpc) is comparable to their own sampling noise: no estimator differs from the reference fit by more than 1.6 times the noise of that difference. Finally, the weighted method is validated against the published Megamaser Cosmology Project result, which it reproduces to within 0.2σ.

1 Introduction

Hubble's Law is one of the foundational results of modern cosmology: it provided the first empirical evidence that the Universe is expanding, and remains central to how astronomers measure cosmic distances, ages, and the overall scale of the Universe. The value of the Hubble constant H_0 sets the present-day expansion rate of the Universe and is directly linked to its estimated age: a higher H_0 implies a younger Universe, and vice versa. Hubble's Law states that the recession velocity v of a galaxy is proportional to its distance r from us:

$$v = H_0\, r \tag{1}$$

where H_0 is the Hubble constant, typically expressed in km/s/Mpc.

It is worth being explicit about what Equation 1 actually is: the leading-order term of the relation between distance and redshift z, valid for $z \ll 1$. Two details matter at the few-per-cent level reached in this project. First, "distance" can mean different things. A distance obtained by comparing the apparent and intrinsic brightness of a standard candle (Cepheids, the tip of the red giant branch, surface brightness fluctuations, supernovae) is a *luminosity distance* D_L; a distance obtained by comparing the physical and angular size of an object (the maser disks of Section 3.1) is an *angular-diameter distance* D_A; the two differ by a factor $(1+z)^2$. Second, to the next order the relation gains a term involving the deceleration parameter q_0:

$$H_0 D_L \approx cz\left[1 + \tfrac{1}{2}(1 - q_0)\, z\right], \qquad H_0 D_A \approx cz\left[1 - \tfrac{1}{2}(3 + q_0)\, z\right]. \tag{2}$$

Every object in both samples has $v \ll c$ (at most $z \approx 0.034$, for the maser galaxy NGC 6264), but for that galaxy the correction to D_A is still about 4%, larger than the statistical error of the final result. For this reason the analysis of the modern sample converts every distance to comoving distance, for which Equation 1 holds exactly (Section 3.2); no attempt is made to constrain q_0 itself from this data.

This project revisits the very first measurement of H_0, published by Edwin Hubble in 1929 [Hubble, 1929], and compares it with a modern estimate obtained using the same statistical

method but with contemporary distance measurements. The goal is not simply to reproduce a historical number, but to understand *why* Hubble's original estimate differed so substantially from the value accepted today, and what this reveals about the evolution of observational astronomy over the past century. Hubble's own initial value, $H_0 \approx 500$ km/s/Mpc, was already revised upward within a few years, and in 1935 Arthur Eddington, combining relativity and quantum theory, confidently claimed to have found "a solution which I regard as definitive," with $H_0 = 865$ km/s/Mpc [Tully, 2023] — a reminder that overconfidence in a single method has a long history in this field, one this project tries to avoid by explicitly comparing two independent approaches, attaching uncertainties to every result, and being transparent about their respective limitations. Despite nearly a hundred years of increasingly precise measurements, the exact value of H_0 remains a subject of active debate. Measurements based on the local Universe — using distance-ladder methods such as Cepheids and Type Ia supernovae — give $H_0 = 73.04 \pm 1.04$ km/s/Mpc [Riess et al., 2022], while measurements inferred from the cosmic microwave background under the standard ΛCDM model give $H_0 = 67.4 \pm 0.5$ km/s/Mpc [Planck Collaboration, 2020]. This discrepancy, known as the *Hubble tension*, is quoted by the SH0ES team at about 5σ significance — meaning that the probability of it being merely a statistical fluctuation would be about 1 in 3.5 million. In 2026 a community-wide combination of many local distance indicators, the Local Distance Network, found $H_0 = 73.50 \pm 0.81$ km/s/Mpc, 7.1σ from the early-Universe value obtained by combining Planck with ground-based CMB data under ΛCDM [H0DN Collaboration, 2026]. Whether the tension signals new physics beyond ΛCDM or unrecognized systematic errors in the local distance ladder is still debated. The most recent analysis of the Chicago-Carnegie Hubble Program (CCHP), which calibrates Type Ia supernovae with the Tip of the Red Giant Branch (TRGB) using both the Hubble and the James Webb Space Telescopes, finds an intermediate value, $H_0 = 70.39 \pm 1.22\ (\text{stat}) \pm 1.33\ (\text{sys}) \pm 0.70\ (\sigma_{\text{SN}})$ km/s/Mpc, which its authors describe as consistent with ΛCDM without the need for new physics [Freedman et al., 2025], a conclusion that remains disputed. While a full treatment of this tension is beyond the scope of this project, it highlights that the precise determination of H_0 remains an open problem at the forefront of astrophysics — one to which this project's own estimates, despite their modest scale, can be compared, provided their uncertainties are stated honestly.

2 Part 1: Reproducing Hubble's Original Analysis (1929)

2.1 Data and Method

I used the original dataset published by Hubble (1929), consisting of 24 extragalactic objects with measured distance r (in megaparsecs) and radial velocity v (in km/s). To estimate the Hubble constant, I first performed a linear fit of the form $v = H_0 r$, using two approaches: an unconstrained linear regression (allowing a free intercept) and a regression forced through the origin, which is the physically motivated form of Hubble's Law, since a galaxy at zero distance should show zero recession velocity due to cosmic expansion. For every fit, I also report the standard error on H_0 from ordinary least squares and the coefficient of determination R^2, to quantify both the precision and the quality of the fit. Hubble's velocities, however, are heliocentric: they include the motion of the Sun itself, which Hubble solved for together with the expansion. Section 2.3 therefore also reproduces his own model, for which the sky coordinates of the 24 objects were added (J2000, from the OpenNGC catalogue for the NGC objects, Verga n.d., and from SIMBAD for the Magellanic Clouds, Wenger et al. 2000). Table 1 reports the full dataset used, taken directly from Hubble's original 1929 paper: distance r (in Mpc) and radial velocity v (in km/s) for each of the 24 objects in the sample.

Object	r (Mpc)	v (km/s)
S. Mag.	0.032	+170
L. Mag.	0.034	+290

Object	r (Mpc)	v (km/s)
N.G.C. 6822	0.214	−130
N.G.C. 598	0.263	−70
N.G.C. 221	0.275	−185
N.G.C. 224	0.275	−220
N.G.C. 5457	0.45	+200
N.G.C. 4736	0.5	+290
N.G.C. 5194	0.5	+270
N.G.C. 4449	0.63	+200
N.G.C. 4214	0.8	+300
N.G.C. 3031	0.9	−30
N.G.C. 3627	0.9	+650
N.G.C. 4826	0.9	+150
N.G.C. 5236	0.9	+500
N.G.C. 1068	1.0	+920
N.G.C. 5055	1.1	+450
N.G.C. 7331	1.1	+500
N.G.C. 4258	1.4	+500
N.G.C. 4151	1.7	+960
N.G.C. 4382	2.0	+500
N.G.C. 4472	2.0	+850
N.G.C. 4486	2.0	+800
N.G.C. 4649	2.0	+1090

Table 1: Distance and radial velocity data for the 24 objects in Hubble's (1929) sample.

2.2 Results

Sample	Free intercept	Forced through origin
Full sample (24 galaxies)	454 ± 75 $(R^2 = 0.62)$	424 ± 42 $(R^2 = 0.62)$
Excluding Local Group (18 galaxies)	409 ± 98 $(R^2 = 0.52)$	433 ± 41 $(R^2 = 0.52)$

Table 2: Estimated H_0 (km/s/Mpc) from Hubble's 1929 data under different fitting assumptions, with standard errors and R^2 from ordinary least squares.

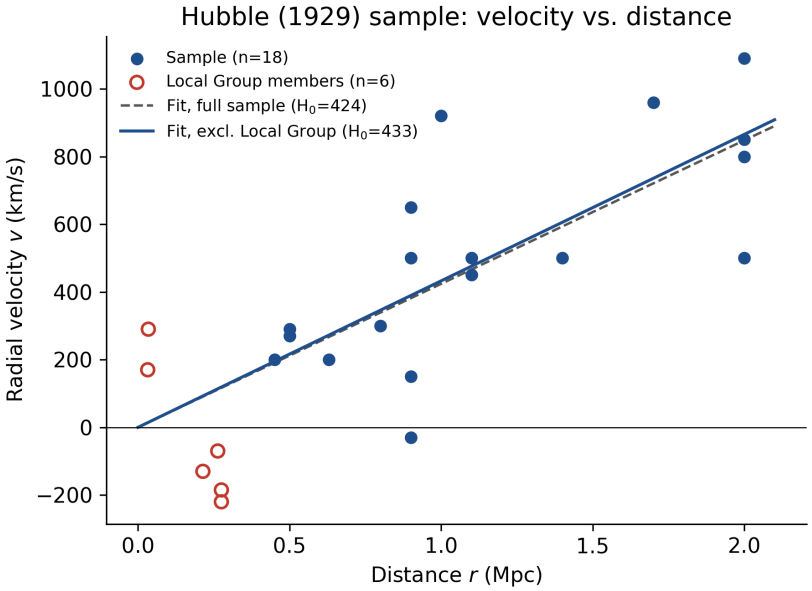


Figure 1: Velocity vs. distance for Hubble's 1929 sample, with linear fits forced through the origin for the full sample and without the Local Group.

For comparison, the currently accepted value of H_0 is approximately 67–73 km/s/Mpc [Riess et al., 2022, Planck Collaboration, 2020], meaning Hubble's original estimate overstated the

expansion rate by a factor of roughly six, a discrepancy of more than three standard errors even for the least precise of the four fits.

2.3 Hubble's own model: the motion of the Sun

The fits above differ from Hubble's in one respect. Hubble did not fit Equation 1 alone: since the velocities of the nearest objects were of the same order as the motion of the Sun, he solved for the expansion term K and for the three components (X, Y, Z) of the solar motion together, through the equations of condition [Hubble, 1929]

$$v = Kr + X\cos\alpha\cos\delta + Y\sin\alpha\cos\delta + Z\sin\delta, \tag{3}$$

where α and δ are the right ascension and declination of each object. Solving Equation 3 by least squares on the same 24 objects gives the results in Table 3.

	This work	Hubble [1929]
K (km/s/Mpc)	465.2 ± 55.7	465 ± 50
X, Y, Z (km/s)	-69, $+235$, -200	-65, $+226$, -195
Solar speed V_0 (km/s)	317	306
Apex (A, D)	$(286°, +39°)$	$(286°, +40°)$
Mean absolute residual (km/s)	150	150

Table 3: Hubble's solution with the solar motion (Equation 3), 24 objects. The small differences in X, Y, Z are expected, since the coordinates used here refer to the equinox of 2000 and Hubble's to that of 1900; the precession between the two rotates (X, Y, Z) but leaves K unchanged.

The fit recovers Hubble's published expansion term, the direction and speed of the solar motion, and his statement that the "residuals average 150 km./sec." The small difference in the quoted uncertainty of K (55.7 against 50) was not investigated. Excluding the Local Group, the same model gives $K = 499 \pm 68$ km/s/Mpc. The difference between the 424 km/s/Mpc of the simple fit and Hubble's 465–500 is therefore entirely due to the solar motion, about 300 km/s, comparable to the velocities of the nearest objects themselves. Hubble also noted that the difference between his two solutions (465 for the 24 objects individually, 513 ± 60 for nine groups) was "due largely to the four Virgo nebulae, which, being the most distant objects and all sharing the peculiar motion of the cluster, unduly influence the value of K" [Hubble, 1929]: the same Virgo Cluster effect reappears in the modern nearby sample (Section 3.5).

2.4 Discussion

Several of the nearest objects in the sample — the Magellanic Clouds, NGC 6822, M33, M32, and M31 (Andromeda) — belong to the Local Group, the small group of galaxies gravitationally bound to the Milky Way, where gravitational interactions between galaxies (peculiar motion) dominate over the recession caused by cosmic expansion. Some velocities in Table 1 are negative: M31 (-220 km/s) and NGC 3031 (M81, -30 km/s). Much of this, however, is not the motion of the galaxies themselves but the reflex of the Sun's own orbit around the Milky Way, at about 250 km/s [Schönrich et al., 2010, McMillan, 2011]. Along the line of sight to M31 this contributes $+192$ km/s, so that M31's modern heliocentric velocity of -301 km/s corresponds to a Galactocentric radial velocity of -109 km/s [van der Marel et al., 2012]: M31 really is approaching the Milky Way, a genuine Local Group effect, but at about a third of its heliocentric speed. For NGC 3031, which belongs to the M81 group outside the Local Group, the reflex is $+115$ km/s, and Hubble's -30 km/s becomes a recession of about $+85$ km/s. This is precisely why Hubble's equations of condition included the solar motion.

Excluding the six Local Group members shifts the two simple fits in opposite directions: the free-intercept slope decreases (454 → 409 km/s/Mpc), while the origin-forced slope increases slightly (424 → 433 km/s/Mpc). Both shifts are comparable to, or smaller than, one standard error on the respective fit, so this asymmetry is only mildly suggestive rather than a strong effect; it is nonetheless consistent with the origin-forced fit being less sensitive to the distortion introduced by nearby peculiar motions, and is therefore treated here as the more reliable estimator in this context. Notably, even after removing the Local Group, H_0 remains around 409–433 km/s/Mpc (499 with Hubble's model) — still roughly six times higher than the modern value, and far outside the uncertainty of every fit. This indicates that neither peculiar motions nor the motion of the Sun can account for Hubble's overestimate. The dominant source of error was a systematic underestimate of the distances themselves, which the following decades corrected in several steps rather than all at once.

It helps first to recall how Hubble's distances were obtained [Hubble, 1929]. Only the first seven objects — the six Local Group members and NGC 5457 — had distances based on Cepheids and other recognizable stars. The next thirteen assumed that the brightest stars in every galaxy share a common upper limit of luminosity, $M_{\mathrm{pg}} \approx -6.3$, calibrated on the nearby galaxies; the last four, members of the Virgo Cluster, were placed at a common distance of about 2 Mpc. Of the 18 objects that remain after excluding the Local Group, therefore, only NGC 5457 has a Cepheid-based distance.

The first correction came from Walter Baade's discovery of two populations of Cepheids with different luminosities [Baade, 1956]: the Cepheids in Hubble's calibrating galaxies are about 1.5 mag brighter, at a given period, than the calibration assumed, which increased extragalactic distances by about a factor of 2 [Tully, 2023] and would therefore roughly halve H_0. Because the brightest-star criterion was calibrated on the same nearby galaxies, this correction propagated to the whole sample — but it accounts for only about a factor of 2 out of the factor of six. The second step came from Humason et al. [1956], who showed that many of the objects taken as the "brightest stars" in more distant galaxies were in fact HII regions, clouds of ionized gas around groups of hot stars, intrinsically much brighter than any single star; with this and the Cepheid correction they obtained $H_0 \approx 180$ km/s/Mpc. Two years later, Sandage [1958] revised the brightest-star calibration further and placed H_0 between about 50 and 100 km/s/Mpc, with a preferred value of 75 [Tully, 2023]. Hubble's factor of six is therefore the product of a chain of errors in the distance scale — the Cepheid zero point first, then the misidentification of HII regions as stars and the recalibration of the brightest-star method — and not of the Cepheid calibration alone. Unlike the statistical uncertainties quoted above, none of these errors can be corrected after the fact within this analysis, since they affect the distances themselves.

2.5 Partial Conclusion

Hubble's 1929 estimate of H_0 was overstated primarily because his distances were systematically too small, rather than because of the specific choice of galaxies in the sample or of the fitting method: with his own model, which includes the motion of the Sun, his published value is reproduced exactly. The underestimate came from a chain of errors in the distance scale of the time: the Cepheid zero point, corrected by Baade, accounts for about a factor of 2, and most of the remainder came from the later revision of the brightest-star distances, which were the majority of the sample, beginning with the recognition that many of Hubble's "brightest stars" were HII regions. This sets up a natural comparison with modern distance measurements in Part 2, using the NED catalogue [NED, n.d.].

3 Part 2: Modern Distance Estimates (NED)

3.1 Data and Method

To compare Hubble's 1929 result with a modern estimate obtained using the same statistical approach, a sample of 39 galaxies outside the Local Group was selected, using distance and redshift data retrieved from the NASA/IPAC Extragalactic Database (NED). For each galaxy, the distance r (in Mpc) was taken from the "Distances" entry, preferring Cepheid-based measurements where available. For galaxies without Cepheid data (mostly ellipticals), Surface Brightness Fluctuations (SBF) were used instead, and for the most distant part of the sample, geometric maser distances from the Megamaser Cosmology Project (MCP) [Pesce et al., 2020] were used, together with one Type Ia supernova (SNIa), one Type II supernova (SNII), and one Tip of the Red Giant Branch (TRGB) distance where no better method was available. These methods, together with their respective systematic uncertainties, are reviewed in detail in Freedman and Madore [2010]. Local Group members were deliberately excluded, applying the same methodological lesson drawn from Part 1. The sample was deliberately extended to include five MCP maser galaxies (UGC 3789, NGC 6264, NGC 5765b, NGC 6323, and CGCG 074-064), which reach distances of 51.5–132.1 Mpc and provide some of the most precise, geometric distance measurements available, independent of the traditional distance ladder. Together with NGC 4258, whose geometric maser distance anchors the distance ladder, they form the six maser-distance galaxies in the sample. NED lists several distance estimates for NGC 4258, including values of about 6.4 and 7.6 Mpc; the maser distance of 7.58 Mpc [Reid et al., 2019] was adopted, as the most precise available and the one on which modern Cepheid calibrations are anchored. The same criterion — adopting the most recent and precise published determination — was applied to the five MCP galaxies, whose distances are taken from the combined MCP analysis [Pesce et al., 2020]. That analysis re-modelled four of them (UGC 3789, NGC 6264, NGC 6323, and NGC 5765b) with revised position-error floors, superseding the earlier individual determinations (49.6, 144, 107, and about 126 Mpc respectively), and it also lists a geometric maser distance for CGCG 074-064 (87.6 Mpc), used here instead of the Fundamental Plane distance (74.4 Mpc) adopted in earlier versions of this project. For transparency, this update alone raised the origin-forced estimates of version 1.2.0 by 2.4 and 2.5 km/s/Mpc relative to version 1.1.0.

Five further distances were corrected in version 1.2.1 after checking them against the literature. NGC 4594 (the Sombrero galaxy) had been listed as a Cepheid distance of 6.22 Mpc, but no Cepheid distance exists for this galaxy: the value matches the lowest globular-cluster luminosity function estimate in the published range (6.22–21.7 Mpc; McQuinn et al. 2016), and it is replaced by the TRGB distance of 9.55 ± 0.13 (stat) $\pm$ 0.31 (sys) Mpc [McQuinn et al., 2016]. For NGC 300, the adopted 1.49 Mpc was well below the Cepheid measurements, and it is replaced by the near-infrared Cepheid distance of 1.88 Mpc ($\mu_0 = 26.37$) of Gieren et al. [2005]. For NGC 2403, NGC 3351, and NGC 3627, the adopted distances (2.88, 8.99, and 8.73 Mpc) were 0.08–0.18 mag shorter than the final HST Key Project Cepheid moduli, 27.48, 29.85, and 29.86 mag (without metallicity correction; Freedman et al. 2001), and are replaced by the corresponding 3.13, 9.33, and 9.38 Mpc. Because these galaxies are all within 10 Mpc, where the fit gives them little weight, the five corrections changed the unweighted full-sample estimates by at most 0.22 km/s/Mpc; the remaining Cepheid distances were not re-sourced (Section 3.6). Table 4 reports the full sample used.

Two velocity frames were tested: the heliocentric radial velocity v_{helio}, and the velocity corrected for the Sun's motion relative to the cosmic microwave background, v_{CMB}, both taken from NED. The exception are the six maser galaxies, for which the systemic velocities measured from the maser disks themselves are used (Table 1 of Pesce et al. 2020, CMB frame), the same values used in the validation of Section 3.4.3; NED's velocities for these galaxies differ from them by up to 140 km/s (NGC 6323). Their heliocentric velocities are obtained by applying, for each galaxy, NED's own difference between the two frames. As in Part 1, two linear fits of the form $v = H_0 r$ were performed for each velocity frame: one with a free intercept, and one

forced through the origin, again with standard errors and R^2 computed for each.

Object	r (Mpc)	Method	$v_{\rm helio}$ (km/s)	$v_{\rm CMB}$ (km/s)	v source
NGC 4258	7.58	Maser	473.3	679.3	Pesce 2020
NGC 3368	9.72	Cepheids	888	1237	NED
NGC 3627	9.38	Cepheids	722	1070	NED
NGC 4321	14.2	Cepheids	1571	1896	NED
NGC 4535	14.8	Cepheids	1964	2299	NED
NGC 4536	12.2	Cepheids	1808	2151	NED
NGC 4548	14.3	Cepheids	486	808	NED
NGC 1365	17.2	Cepheids	1636	1539	NED
NGC 1425	20.9	Cepheids	1512	1412	NED
NGC 2403	3.13	Cepheids	133	182	NED
NGC 300	1.88	Cepheids	147	−88	NED
NGC 3351	9.33	Cepheids	779	1128	NED
NGC 3982	20.5	Cepheids	1122	1291	NED
NGC 4496A	13.6	Cepheids	1730	2072	NED
NGC 4639	19.7	Cepheids	1025	1345	NED
IC 4182	4.07	Cepheids	321	550	NED
NGC 5253	3.01	Cepheids	407	681	NED
NGC 7331	14.4	Cepheids	816	490	NED
NGC 4038	18.1	Cepheids	1624	1979	NED
NGC 5236	4.5	Cepheids	513	793	NED
NGC 4594	9.55	TRGB	1089	1433	NED
NGC 1097	24.1	SNII (optical)	1271	1105	NED
NGC 2841	13.1	Cepheids	635	806	NED
NGC 3379	8.8	SBF	907	1255	NED
NGC 3115	8.51	SBF	686	1041	NED
NGC 1023	9.6	SBF	636	432	NED
NGC 4494	15	SBF	1342	1634	NED
NGC 5866	14.7	SBF	755	825	NED
NGC 4278	14.9	SBF	649	934	NED
NGC 7619	53	SBF	3762	3391	NED
NGC 3384	9.46	SBF	704	1051	NED
NGC 4589	20.4	SBF	1980	2030	NED
NGC 2986	29.7	SNIa	2302	2637	NED
NGC 1400	16.3	SBF	590	467	NED
UGC 3789	51.5	Maser	3259.9	3319.9	Pesce 2020
NGC 6264	132.1	Maser	10192.6	10192.6	Pesce 2020
NGC 5765b	112.2	Maser	8317.7	8525.7	Pesce 2020
NGC 6323	109.4	Maser	7834.5	7801.5	Pesce 2020
CGCG 074-064	87.6	Maser	6912.2	7172.2	Pesce 2020

Table 4: Distance (as published: angular-diameter distance for the masers, luminosity distance for the others), measurement method, heliocentric and CMB-frame velocity for the modern galaxy sample (Local Group excluded, $n = 39$). Distances of the maser galaxies are from Reid et al. [2019] (NGC 4258) and Pesce et al. [2020] (the five MCP galaxies), those of NGC 300, NGC 2403, NGC 3351, NGC 3627, and NGC 4594 from Gieren et al. [2005], Freedman et al. [2001], and McQuinn et al. [2016] (see text); all other values are from NED.

3.2 From distances and redshifts to the Hubble law

Two kinds of distance enter the sample. Cepheid, TRGB, SBF and supernova distances come from distance moduli and are luminosity distances D_L. Maser distances come from the ratio between the physical and the angular size of the disk and are angular-diameter distances D_A; Pesce et al. [2020] relate them to redshift accordingly (their Equation 1). In an expanding

universe, Equation 1 is exact for neither of them, but for the comoving distance D_C, provided the velocity is understood as

$$H_0 D_C = c \int_0^z \frac{dz'}{E(z')}, \qquad E(z) = \sqrt{\Omega_m (1+z)^3 + 1 - \Omega_m}, \tag{4}$$

in a flat ΛCDM universe, with $z = v/c$ and $\Omega_m = 0.315$ [Planck Collaboration, 2020], as in Pesce et al. [2020]. The distances convert as

$$D_C = \frac{D_L}{1+z} = D_A\,(1+z), \tag{5}$$

and for $z \ll 1$ the right-hand side of Equation 4 is $cz\left[1 - \frac{1}{2}(1+q_0)z\right]$, with $q_0 = \frac{3}{2}\Omega_m - 1 \approx -0.53$. The factor $(1+z)$ in Equation 5 depends on no cosmological model; only the much smaller term in q_0 does, and changing Ω_m from 0.25 to 0.35 moves the results below by at most 0.1 km/s/Mpc. All distances are converted with the CMB-frame redshift. Table 5 lists the effect on the seven galaxies beyond 25 Mpc; for the other 32 it is at most 0.6%.

Galaxy	Method	z_{CMB}	v_{CMB}/r as measured	after conversion
NGC 7619	SBF	0.0113	63.98	64.53 (+0.86%)
NGC 2986	SNIa	0.0088	88.79	89.38 (+0.67%)
UGC 3789	Maser	0.0111	64.46	63.59 (−1.35%)
NGC 6264	Maser	0.0340	77.16	74.02 (−4.07%)
NGC 5765b	Maser	0.0284	75.99	73.39 (−3.42%)
NGC 6323	Maser	0.0260	71.31	69.07 (−3.14%)
CGCG 074-064	Maser	0.0239	81.87	79.51 (−2.89%)

Table 5: Effect of the conversion to comoving distance on v_{CMB}/r (km/s/Mpc) for the galaxies beyond 25 Mpc.

The changes are small for every single galaxy, but not for the fit. In an origin-forced least-squares fit each galaxy's weight is proportional to r^2, so the six masers carry 84.5% of the total weight and the seven galaxies beyond 25 Mpc 90.4%; equivalently, the effective number of galaxies, $(\sum r^2)^2/\sum r^4$, is only 5.6. Without the conversion, the origin-forced fits of the next section would give 74.6 km/s/Mpc (v_{helio}) and 76.2 km/s/Mpc (v_{CMB}), about 2.1 km/s/Mpc higher, which is more than their statistical error; the factor $(1+z)$ accounts for about 1.7 of this, the q_0 term for 0.4. Versions up to 1.2.1 of this analysis fitted the measured distances and velocities directly. All results below, figures included, use the comoving quantities.

3.3 Results

Fit type	H_0, v_{helio} (km/s/Mpc)	H_0, v_{CMB} (km/s/Mpc)
Free intercept	71.3 ± 1.9 $(R^2 = 0.974)$	70.0 ± 2.5 $(R^2 = 0.954)$
Forced through origin	72.5 ± 1.5 $(R^2 = 0.973)$	74.1 ± 2.1 $(R^2 = 0.945)$

Table 6: Estimated H_0 from the extended modern sample ($n = 39$, Local Group excluded, comoving distances), using heliocentric vs. CMB-frame velocities, with standard errors and R^2 from ordinary least squares.

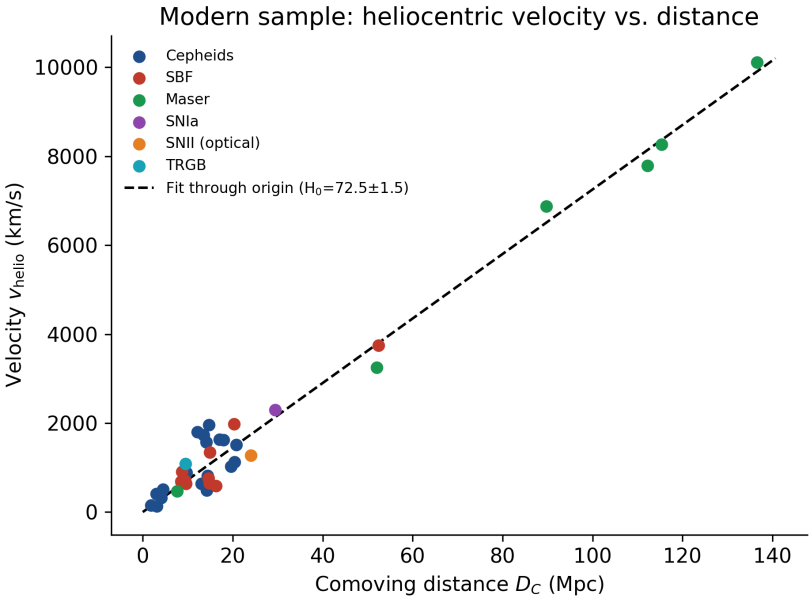


Figure 2: Heliocentric velocity vs. comoving distance for the full modern sample ($n = 39$), including the Megamaser Cosmology Project galaxies out to 132 Mpc (137 Mpc in comoving distance), with linear fit.

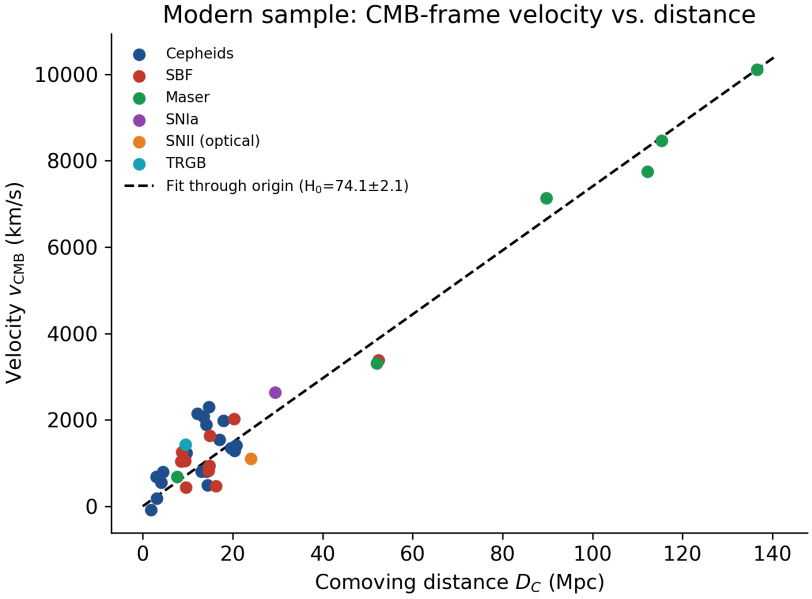


Figure 3: CMB-frame velocity vs. comoving distance for the full modern sample ($n = 39$), with linear fit.

3.4 Weighted Fit

The fits above treat every galaxy as equally informative, regardless of how precisely its distance is known. As a first attempt at correcting this, each galaxy was assigned a representative fractional distance uncertainty based on its distance method, using the typical ranges summarized in Freedman and Madore [2010]: 7% for Cepheids, 10% for SBF, 8% for SNIa, 15% for SNII (optical), and 5% for masers; the single TRGB distance is given 5%, slightly more than its own quoted uncertainty of about 3.5% [McQuinn et al., 2016]. These are method-level, representative values, not individual NED-quoted errors for each object, and should be read as an order-of-magnitude correction rather than a precise error budget.

A first, naive attempt is instructive precisely because it fails: weighting each point by the distance-implied uncertainty on v alone ($\sigma_v = H_0 \cdot f_{\rm method} \cdot r$, propagated through an iteratively reweighted least-squares scheme) pulls the origin-forced fit to $H_0 = 80.6 \pm 4.4$ km/s/Mpc for $v_{\rm helio}$ and 94.8 ± 7.8 km/s/Mpc for $v_{\rm CMB}$ — both well outside the accepted range, with a poor fit quality ($\chi^2/{\rm dof} = 24$ and 55 respectively). The reason is instructive: because $\sigma_v \propto r$ in this naive scheme, nearby galaxies are assigned artificially small uncertainties and therefore dominate the fit, even though it is precisely at small r that peculiar motions (not distance-measurement error) are the dominant source of scatter. In effect, naive distance-only weighting reintroduces the same low-r, peculiar-velocity-dominated bias that the sample extension described in Section 3.1 was designed to suppress.

A more defensible weighting scheme therefore adds a distance-independent peculiar-velocity floor $\sigma_{\rm pec}$ in quadrature, $\sigma_v = \sqrt{(H_0 \cdot f_{\rm method} \cdot r)^2 + \sigma_{\rm pec}^2}$, reflecting the fact that peculiar velocities of a few hundred km/s do not shrink simply because a galaxy's distance happens to be well measured. Rather than adopting a value for $\sigma_{\rm pec}$, it is estimated from the data, as the value for which the origin-forced weighted fit has $\chi^2/{\rm dof} = 1$: the scatter that the distance errors alone do not explain. This gives $\sigma_{\rm pec} = 368$ km/s for $v_{\rm helio}$ and 528 km/s for $v_{\rm CMB}$; a joint maximum-likelihood fit of H_0 and $\sigma_{\rm pec}$ gives almost the same values (372 and 529 km/s, with $H_0 = 72.61$ and 74.65 km/s/Mpc). Both are larger than the typical galaxy peculiar velocities of order 100–300 km/s relative to the Hubble flow. This is not surprising for this sample: several of its nearby galaxies lie in the region of the Virgo Cluster, whose members move at several hundred km/s relative to the cluster itself, and in the CMB frame the nearby galaxies also carry part of the motion of the Local Group (Section 3.5). With these floors:

Fit type	H_0, $v_{\rm helio}$ (km/s/Mpc)	H_0, $v_{\rm CMB}$ (km/s/Mpc)
Free intercept	70.8 ± 2.6 ($\chi^2/{\rm dof} = 0.99$)	69.6 ± 3.0 ($\chi^2/{\rm dof} = 0.86$)
Forced through origin	72.7 ± 2.1 ($\chi^2/{\rm dof} = 1.00$)	74.8 ± 2.6 ($\chi^2/{\rm dof} = 1.00$)

Table 7: Weighted H_0 estimates using inverse-variance weights combining method-level distance errors with the peculiar-velocity floor estimated from the data ($\sigma_{\rm pec} = 368$ km/s for $v_{\rm helio}$, 528 km/s for $v_{\rm CMB}$).

The weighted origin-forced estimates (72.7 ± 2.1 and 74.8 ± 2.6 km/s/Mpc) are close to the unweighted ones (72.5 ± 1.5 and 74.1 ± 2.1). The choice of $\sigma_{\rm pec}$ matters mainly in the CMB frame: a fixed floor of 150, 200 or 300 km/s gives 73.9, 73.3 and 72.8 km/s/Mpc for $v_{\rm helio}$, but 79.7, 77.8 and 76.0 km/s/Mpc for $v_{\rm CMB}$, with $\chi^2/{\rm dof}$ between 1.4 and 8.8. The fixed floor of 200 km/s used up to version 1.2.1 gives $\chi^2/{\rm dof} = 2.9$ and 5.7: the fit itself indicates that such a floor is too small, and too small a floor gives the nearby galaxies, whose scatter it underestimates, too much weight, which pushes the CMB-frame estimate up. A single isotropic floor remains a simplification, since real peculiar motions are direction-dependent (e.g. infall toward the Virgo Cluster or the Great Attractor) and larger near clusters. The fitted value is set mostly by the nearby galaxies: the seven galaxies beyond 25 Mpc scatter around the reference fit by only about

300 km/s (root mean square), against about 500 km/s expected from the adopted error model.

As a complementary check, the unweighted fractional residuals, $(v - H_0 r)/(H_0 r)$, were also examined by method. Below, each group mean is quoted with its standard error ($s/\sqrt{n}$, where s is the scatter of the group), and the scatter is given separately. The 20 Cepheid-based points have a positive mean residual of $+0.19 \pm 0.10$ (scatter 0.46), 1.9 standard errors from zero: they lie on average above the fit, as expected for nearby galaxies, whose velocities are strongly affected by local peculiar motions — two of the three largest residuals (+0.8 to +1.1) belong to NGC 4535 and NGC 4536, in the region of the Virgo Cluster — and possibly also because some of their distances are still somewhat short (Section 3.6). The 10 SBF points are consistent with zero (-0.01 ± 0.10, scatter 0.31), and the 6 masers are both centered near zero and far tighter than either (-0.04 ± 0.03, scatter 0.08). The single SNIa, SNII, and TRGB points cannot support a statistical statement on their own. This is consistent with, though not strong independent proof of, the geometric maser distances being the most internally precise method in the sample — exactly why Section 3.5 below finds them so influential.

3.4.1 A check on errors in both variables (ODR)

Every fit so far, weighted or not, is a form of ordinary least squares: it assumes the distance r is known exactly and puts all the measurement error on the velocity v. This is not quite right, since r is, if anything, the less certain of the two quantities. Orthogonal Distance Regression (ODR) relaxes this assumption: instead of minimizing vertical distances to the fit line, it minimizes a weighted combination of the horizontal and vertical distances, using an error budget on *both* axes. On the distance axis this is $\sigma_r = f_{\rm method} \cdot r$; on the velocity axis it includes only the terms that genuinely belong to the velocity, namely the fitted peculiar-velocity floor and a representative measurement error of 5 km/s, added in quadrature. For a straight line through the origin, ODR is equivalent to minimizing

$$S(H_0) = \sum_i \frac{(v_i - H_0 r_i)^2}{\sigma_{v,i}^2 + H_0^2\, \sigma_{r,i}^2}, \tag{6}$$

so ODR itself propagates the distance error onto the velocity axis, and the effective variance of each point is the same combination used in the weighted fit. This form is used here; it gives the same results as the `scipy.odr` implementation of earlier versions, which SciPy has deprecated. Version 1.1.0 mistakenly also included the distance term $H_0 f_{\rm method} r$ in σ_v, counting the distance error twice and biasing the ODR estimates upward; this was corrected in version 1.2.0.

The ODR gives $H_0 = 72.9 \pm 2.1$ km/s/Mpc for $v_{\rm helio}$ (compared to 72.7 ± 2.1 for the weighted fit) and $H_0 = 74.9 \pm 2.6$ km/s/Mpc for $v_{\rm CMB}$ (compared to 74.8 ± 2.6). The direction of the change is the expected one: ignoring measurement error on the independent variable causes ordinary least squares to systematically *underestimate* the slope of a line, an effect statisticians call regression dilution or attenuation bias, so allowing for it pushes H_0 up. Here the effect is small (0.2 km/s/Mpc in both frames), well within one standard error.

Monte Carlo check of the estimators. To test the estimators themselves, 1,000 samples were simulated with the same distances and distance methods as the real one, a true H_0 of 72 km/s/Mpc, Gaussian fractional distance errors $f_{\rm method}$ and Gaussian peculiar velocities of 368 km/s. Ordinary least squares through the origin, the weighted fit and the ODR recover the input value with mean biases of -0.29, -0.15 and $+0.09$ km/s/Mpc respectively: negligible compared with their scatter, and in the order expected from attenuation bias. The scatter of the simulated estimates, 2.0–2.1 km/s/Mpc, is larger than the errors measured on the real data (1.4–1.5 km/s/Mpc for the reference fit, Section 3.5), because a single floor of 368 km/s overstates the scatter of the distant galaxies, as noted above. The errors measured on the real data are therefore the more relevant ones, with the caveat that resampling a sample whose weight is concentrated in about six galaxies is itself approximate.

3.4.2 Statistical error versus choice of method

The five estimators applied to each velocity frame (ordinary least squares with a free intercept or through the origin, the two weighted fits, and ODR) span 70.8–72.9 km/s/Mpc for $v_{\rm helio}$ and 69.6–74.9 km/s/Mpc for $v_{\rm CMB}$. Because all ten are computed from the same 39 galaxies, their differences are partly sampling noise: each estimator weights the same few distant galaxies differently, and so picks up their scatter differently. To separate the two effects, all ten estimators were recomputed on the same 4,000 bootstrap resamples of the sample, including the estimate of the peculiar-velocity floor (Table 8). This gives both the sampling noise of each estimator and the sampling noise of each *difference* between an estimator and the reference fit ($v_{\rm helio}$, ordinary least squares through the origin).

Frame	Estimator	H_0	SE	$\mathrm{SD}_{\rm boot}$	Δ	$\Delta/\mathrm{SD}(\Delta)$
$v_{\rm helio}$	OLS, origin (reference)	72.53	1.52	1.36	—	—
	OLS, free intercept	71.33	1.93	1.79	−1.21	−0.8
	weighted, origin	72.72	2.05	1.71	+0.19	+0.3
	weighted, free intercept	70.80	2.59	1.89	−1.73	−1.2
	ODR	72.90	2.06	1.73	+0.36	+0.6
$v_{\rm CMB}$	OLS, origin	74.06	2.14	1.98	+1.52	+1.4
	OLS, free intercept	69.99	2.53	2.51	−2.55	−1.2
	weighted, origin	74.75	2.59	2.30	+2.22	+1.6
	weighted, free intercept	69.63	2.99	2.57	−2.90	−1.3
	ODR	74.93	2.61	2.37	+2.39	+1.6

Table 8: The ten H_0 estimators of this analysis (km/s/Mpc): analytic standard error (SE), standard deviation over 4,000 paired bootstrap resamples ($\mathrm{SD}_{\rm boot}$; the peculiar-velocity floor is re-estimated in every resample), shift from the reference fit (Δ), and that shift in units of its own bootstrap standard deviation.

No estimator differs from the reference by more than 1.6 times the sampling noise of that difference. With 39 galaxies, therefore, the dependence of H_0 on the choice of estimator cannot be distinguished from sampling noise: the spread across estimators is best read as a visible symptom of the statistical uncertainty, rather than as an additional systematic error to be added on top of it (doing so would count much of the same noise twice). The heliocentric estimators agree closely (standard deviation 0.9 km/s/Mpc across the five), while the CMB-frame ones are more spread out (2.6 km/s/Mpc), consistent with the frame comparison in the Discussion below: in the CMB frame the nearby galaxies scatter much more, and the estimators that give them more weight move further.

3.4.3 Validation against the published MCP result

As an external check of the method itself, the weighted origin-forced fit of Section 3.4 was applied to the six MCP galaxies using the distances, distance uncertainties (symmetrized) and CMB-frame velocities published in Table 1 of Pesce et al. [2020], converted to comoving quantities as in Section 3.2, with the same fixed 250 km/s peculiar-velocity uncertainty adopted there. It gives $H_0 = 73.1 \pm 3.0$ km/s/Mpc, compared with the published 73.9 ± 3.0 km/s/Mpc: a difference of 0.2σ. Without the conversion, the same fit gives 75.4 ± 3.1 km/s/Mpc, 0.4σ above the published value. Versions up to 1.2.1 attributed that difference to the asymmetric distance likelihoods of the published analysis, but most of it came from treating the angular-diameter distances as if they were comoving distances, which Pesce et al. [2020] account for in their Equation 1. The small remaining difference is expected from the symmetrized errors. Within its precision, the simple method used throughout this project reproduces a professional result.

3.5 Discussion

The four unweighted modern estimates (70.0–74.1 km/s/Mpc) lie within the range spanned by current measurements, $H_0 \approx 67$–74 km/s/Mpc [Planck Collaboration, 2020, Riess et al., 2022, H0DN Collaboration, 2026], toward its upper end, in stark contrast with Hubble's 1929 results (409–454 km/s/Mpc, depending on fit type and sample). Since the fitting procedure applied here is identical to the one used in Part 1, this large improvement cannot be attributed to a change in statistical method: it must instead reflect the far greater accuracy of modern distance measurements compared to those available in 1929. As in Part 1, the origin-forced and free-intercept fits differ by less than two standard errors in both velocity frames, the origin-forced fit lying slightly above; the high R^2 values (0.95–0.97) show that a single linear relation accounts for the great majority of the variance in the velocities. The most notable result of this extension concerns the relationship between the two velocity frames. Repeating the same fits on the nearby galaxies alone ($r \leq 25$ Mpc, $n = 32$) shows that there, switching from heliocentric to CMB-frame velocities *worsens* the estimate: the origin-forced fit moves from 75.4 ± 5.2 to 85.9 ± 7.1 km/s/Mpc, and the free-intercept fit from 62.0 ± 12.6 to 55.0 ± 16.2 km/s/Mpc. These nearby-only fits are also far less precise ($R^2 = 0.17$–0.45, against 0.95–0.97 for the full sample), because at these distances peculiar velocities of a few hundred km/s are comparable to the expansion signal itself. Once the seven galaxies beyond 25 Mpc are added (the five MCP masers, at 51.5–132.1 Mpc, together with NGC 7619 and NGC 2986), this discrepancy nearly disappears: the heliocentric and CMB-frame fits now agree to within about 1.5 km/s/Mpc (71.3 vs. 70.0 for the free intercept; 72.5 vs. 74.1 forced through the origin). A likely explanation is that the CMB frame is the wrong reference for nearby galaxies. The correction to the CMB frame removes the Sun's motion with respect to the CMB, most of which is the motion of the Local Group as a whole; the nearby galaxies largely share that motion, being carried by the same large-scale flows, in addition to the more local pull of the Virgo Cluster, so for them the correction adds a velocity they do not have relative to us. In the true Hubble flow these motions become comparatively negligible, and the two frames agree. Testing this directly would require velocities in the Local Group frame or corrected with a model of the local flows (Section 3.6). Hubble himself had already noted that the Virgo galaxies, sharing the cluster's motion, unduly influenced his value of K (Section 2.3).

How much do the six maser galaxies matter? Because the argument above rests on extending the sample to large distances, and because the five MCP galaxies (at 51.5–132.1 Mpc) are the points that reach farthest into the Hubble flow, it is worth asking directly how much the maser-distance galaxies, specifically, are driving the result. Repeating the origin-forced fit with all six of them removed (NGC 4258 included; $n = 33$) gives $H_0 = 74.6 \pm 4.1$ km/s/Mpc for $v_{\rm helio}$ and $H_0 = 80.0 \pm 5.8$ km/s/Mpc for $v_{\rm CMB}$. Removing them therefore raises the central value by 2.1 and 5.9 km/s/Mpc respectively, 0.5 and 1.0 times the standard error of the maser-free fit, and increases the uncertainty by a factor of 2.7. The masers set the precision of the result, by providing a long, precise lever arm; they also pull its central value down, but by no more than about one standard error of the fit without them. Earlier versions of this project (up to 1.1.0) found that removing the masers raised H_0 by about 10 km/s/Mpc; that shift came largely from CGCG 074-064, which with its Fundamental Plane distance of 74.4 Mpc implied $v/r \approx 93$ km/s/Mpc, whereas it is now counted, with its geometric distance, among the maser galaxies. A complementary leave-one-out (jackknife) test, in which each galaxy is removed one at a time, shows that no single galaxy matters much: removing any one of them shifts the origin-forced H_0 by at most 0.8 km/s/Mpc for $v_{\rm helio}$ and 1.2 km/s/Mpc for $v_{\rm CMB}$ (NGC 6323 in both cases). The jackknife standard error (1.3 km/s/Mpc for $v_{\rm helio}$, 1.7 for $v_{\rm CMB}$) and the standard deviation of a bootstrap resampling of the full sample (20,000 resamples; 1.4 and 2.0 km/s/Mpc) agree reasonably with the analytic standard errors in Table 6 (1.5 and 2.1 km/s/Mpc), and the bootstrap 68% confidence intervals are $[71.3, 73.7]$ km/s/Mpc for $v_{\rm helio}$

and [72.6, 75.9] km/s/Mpc for $v_{\rm CMB}$. Taken together, these checks indicate that the result is a property of the extended sample as a whole — specifically, of reaching into the true Hubble flow — rather than an artifact of one favorably-placed data point.

As a further check, the case of M31 (Andromeda) was examined with its modern velocity. Its heliocentric velocity, −301 km/s, is still negative, as in Hubble's data, but as discussed in Part 1 about two thirds of it is the reflex of the Sun's orbit in the Milky Way: in the Galactocentric frame M31 approaches at −109 km/s [van der Marel et al., 2012]. The approach itself is a genuine dynamical effect of the Local Group, where the mutual gravity of the Milky Way and M31 dominates over cosmic expansion — the same kind of effect, on a larger scale, as the peculiar velocities that distort the nearby subsample above.

Finally, Figure 4 compares the origin-forced estimates with the literature. Using statistical errors only (the bootstrap standard deviation of each estimator combined with the quoted literature error), the reference estimate, 72.5 ± 1.5 km/s/Mpc, lies 0.3σ from the SH0ES value [Riess et al., 2022], 0.6σ from the Local Distance Network [H0DN Collaboration, 2026], 0.9σ from the CCHP TRGB value [Freedman et al., 2025], and 3.5σ above the Planck value [Planck Collaboration, 2020]. Across all ten estimators of Table 8, the distance from SH0ES ranges from -1.2σ to $+0.7\sigma$ and from the Local Distance Network from -1.4σ to $+0.6\sigma$, while the distance from Planck ranges from 0.9σ to 3.5σ; the four free-intercept variants lie between 0.9σ and 2.1σ from Planck. This places the project's result on the local side of the tension, consistent with all three local measurements but unable to distinguish between them: the CCHP and SH0ES values differ by 2.7 km/s/Mpc, more than the statistical error of this estimate. Nor should the result be read as independent support for the local side. The measurement is built entirely on local distance indicators. Its most important ingredient, the maser distances, are the same ones from which Pesce et al. [2020] derive their own local value of 73.9 ± 3.0 km/s/Mpc; NGC 4258 is the geometric anchor of the SH0ES Cepheid calibration; and choices about the data have moved the result by amounts comparable to its statistical error: by +2.5 km/s/Mpc when the updated maser distances were adopted (version 1.2.0), and by −2.1 km/s/Mpc when the distances were converted to comoving distance (this version). The comparison therefore reproduces, on a small scale, the same discrepancy between local and early-Universe determinations that defines the tension, rather than shedding light on its cause. It is also a useful reminder that "landing near a published value" and "being a precise, well-controlled measurement" are not the same claim, and this project is more confidently the former than the latter.

3.6 Limitations

Several limitations of this analysis should be acknowledged. First, the galaxy sample was not selected through a systematic or blind procedure, but by prioritizing objects with well-documented, high-quality distance measurements on NED; this could introduce a mild selection bias favoring galaxies that are easier to measure precisely. Second, the weighted fit in Section 3.4 uses representative, method-level fractional errors rather than the individual quoted uncertainty of each galaxy (for the maser galaxies in particular, the single 5% value is a simplification: the individual uncertainties published by Pesce et al. [2020] and Reid et al. [2019] range from about 1.5% for NGC 4258 to about 26% for NGC 6323), and its peculiar-velocity floor, although now estimated from the data, is a single isotropic number for all galaxies: it is set mostly by the nearby galaxies and overstates the scatter of the distant ones. Third, only two velocity frames (heliocentric and CMB) were tested; a more complete treatment would use velocities corrected with a model of the local flows (Virgo Cluster infall, the Great Attractor, and others), rather than relying on sample extension and a scatter floor to suppress their effect. For the maser galaxies alone, Pesce et al. [2020] find that different treatments of peculiar velocities move H_0 between 71.8 and 76.9 km/s/Mpc, so this is a real source of systematic uncertainty. Fourth, as the sensitivity analysis in Section 3.5 makes explicit, the precision of the result depends on a small number of long-baseline galaxies (above all the five MCP masers: the effective number

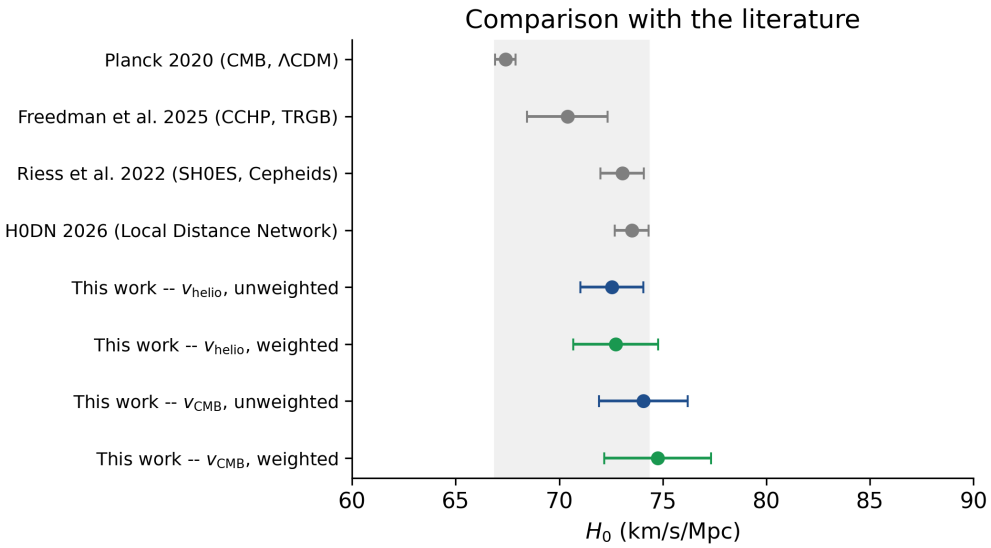


Figure 4: Comparison of this project's origin-forced H_0 estimates (unweighted and weighted, both velocity frames) with Planck 2020 [Planck Collaboration, 2020], the CCHP TRGB value of Freedman et al. 2025 [Freedman et al., 2025], Riess et al. 2022 [Riess et al., 2022] (SH0ES), and the Local Distance Network [H0DN Collaboration, 2026]. Error bars on this project's estimates are the analytic standard errors. The shaded band spans from the lower end of the Planck uncertainty to the upper end of the local distance-ladder values.

of galaxies in the fit is about six), and its central value depends on choices about their data: adopting the distances of Pesce et al. [2020] moved the estimates by about +2.5 km/s/Mpc, and treating them correctly as angular-diameter distances by −2.1 km/s/Mpc. This is not a flaw in the method, but it does mean the modern estimate here is less independent of any single distance-measurement technique than a large professional sample would be. Fifth, the Cepheid distances, taken from NED, do not come from a single homogeneous calibration: the four that were checked against the literature in version 1.2.1 had to be replaced (Section 3.1), and the other 16 have not been re-sourced from a single reference such as the final HST Key Project analysis [Freedman et al., 2001]. Because these galaxies all lie within about 21 Mpc, where the fit gives them little weight, this mainly affects the nearby-subsample fits and the Cepheid residuals rather than the full-sample estimate. Sixth, the conversion to comoving distance assumes a flat ΛCDM cosmology; this matters little, since the model-independent factor $(1+z)$ dominates and changing Ω_m from 0.25 to 0.35 moves the results by at most 0.1 km/s/Mpc. Finally, the sample size ($n = 39$), while sufficient to obtain a stable estimate of H_0, remains small compared to the hundreds or thousands of objects used in professional determinations of the Hubble constant, so the results presented here should be understood as a stable proof-of-concept reproduction of a well-established method rather than a new independent constraint on H_0.

4 Conclusion

Reproducing Hubble's original 1929 analysis — including, as he did, the motion of the Sun — recovers his published value of $K = 465$ km/s/Mpc, an estimate of H_0 roughly six times higher than the modern accepted value, primarily because the distances of the time were systematically too small: the result of a chain of errors, from the Cepheid zero point (about a factor of 2) to the misidentification of HII regions as the brightest stars in more distant galaxies (compounded, to a lesser extent, by the inclusion of nearby galaxies whose peculiar motion is not representative of cosmic expansion). Applying the same statistical method to an extended modern sample of 39

galaxies, spanning distances from about 1.9 Mpc to 132 Mpc and using a combination of Cepheid, SBF, supernova, TRGB and geometric maser distances, yields unweighted H_0 estimates of 70.0–74.1 km/s/Mpc, with statistical errors of 1.5–2.5 km/s/Mpc. The reference fit (heliocentric velocities, through the origin) gives 72.5 ± 1.5 km/s/Mpc: on the local side of the Hubble tension and consistent with all current local measurements, as expected for a measurement built entirely on local distance indicators. Reaching this required a correction that is easy to overlook: although every galaxy has $z < 0.035$, the maser distances are angular-diameter distances, and fitting them directly against redshift would have raised the result by about 2 km/s/Mpc, more than its statistical error. Jackknife and bootstrap resampling support the analytic uncertainties, and show that the six maser-distance galaxies set the precision of the result, by providing a long, geometric lever arm into the Hubble flow, while moving its central value by about one standard error of the fit without them, or less. Inverse-variance weighting, with a peculiar-velocity floor estimated from the data, and orthogonal distance regression move the estimate within 69.6–74.9 km/s/Mpc depending on velocity frame and estimator; a Monte Carlo simulation shows that these estimators are unbiased, and a paired bootstrap shows that their differences are comparable to their own sampling noise (at most 1.6 times the noise of each difference), so with 39 galaxies the choice of method cannot be separated from statistical noise. The weighted method itself reproduces the published MCP value to within 0.2σ. If anything, the most important lesson of the project is that a defensible error budget, not just a best-fit number, is what turns a curve fit into a measurement — and that this includes errors in the analysis itself: successive versions corrected a double counting of the distance uncertainty, five individual distances that were mislabelled or outdated, a fixed peculiar-velocity floor that the data did not support, and the treatment of angular-diameter distances, the last of which alone moved the result by about 2 km/s/Mpc. Taken together, the comparison illustrates that the key advance separating Hubble's original measurement from the modern value of H_0 was not a change in statistical methodology, but nearly a century of improvement in the precision of the extragalactic distance scale — together with the importance of sampling galaxies far enough into the Hubble flow that local peculiar motions no longer bias the result, and of being honest about how much any single analysis choice can still move the answer.

A NED Query Details

Distances and velocities for the modern sample (Table 4) were retrieved from the NASA/IPAC Extragalactic Database (NED) in June 2026. Each object's common designation as listed in Table 4 (NGC, UGC, IC, or CGCG catalogue number) is also its NED object name and can be used directly in NED's "By Name" search to retrieve the same "Basic Data" and "Distances" entries used here, making the dataset independently reproducible without requiring a separate identifier list. The exceptions are the distances and velocities of the six maser galaxies, taken from Reid et al. [2019] (NGC 4258 distance) and Table 1 of Pesce et al. [2020] (the five MCP distances and all six velocities), and five distances corrected after checking them against the literature: NGC 300 [Gieren et al., 2005], NGC 2403, NGC 3351, and NGC 3627 [Freedman et al., 2001], and NGC 4594 [McQuinn et al., 2016]. The file `mcp_pesce2020.csv` in the repository reproduces the distances, distance uncertainties and CMB-frame velocities of that table, used for the validation in Section 3.4.3, and the column `v_source` of `ned_modern.csv` records the source of each velocity. The J2000 coordinates of Hubble's 24 objects, used in Section 2.3, are included in `hubble1929.csv`.

Data Availability

All distance and redshift data used in this analysis are publicly available from the NASA/IPAC Extragalactic Database (NED) at `https://ned.ipac.caltech.edu` (see Appendix A for query

details). Hubble's original 1929 dataset is reproduced from Hubble [1929]. The Python code used for the fits, uncertainty estimates, and sensitivity analysis, together with its unit tests, both datasets and all figures, is openly available at https://github.com/Viserion137/hubble-constant-project and archived on Zenodo; the DOI https://doi.org/10.5281/zenodo.22260280 covers all archived versions.

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